Differentiable Rendering Using Gaussian Mixtures

Geometry, Probability, Computer Graphics

Problem

Rendering synthesizes images from geometric descriptions of a scene. Traditional rendering pipelines operate on primitives such as triangles and surfaces, but their discrete visibility and rasterization operations often make differentiation with respect to scene parameters difficult. Gaussian Mixtures offer an alternative geometric representation, enabling smooth rendering operations and differentiable image synthesis. The project explores geometry-induced Gaussian representations for differentiable rendering. Starting from existing geometric models, such as triangle meshes or parametric surfaces, Gaussian components will be constructed and used for image synthesis. Possible directions include:

- Constructing Gaussian components from geometric primitives.
- Investigating Gaussian means, covariances, and mixture weights.
- Developing differentiable projection, visibility, and compositing.
- Evaluating approximation quality, gradients, and rendering efficiency.

Prerequisites

- Firm grasp over Linear/Multilinear Algebra and Analysis
- Good understanding of Probability and Statistics
- Familiarity with Computer Graphics and geometric representations is advantageous
- PyTorch or an equivalent NumPy-like framework
- Some exposure to Differential Geometry is advantageous
- Interest in numerical methods, mathematical modelling, and computer graphics

References

- Gaussian Mesh Renderer for Lightweight Differentiable Rendering by Liu and Okura

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